Milestone 5: Model Performance Evaluation and Results Analysis

1. Trained Model and Pipeline Overview

The system utilizes a **RoBERTa-base** transformer architecture, specifically a fully fine-tuned model (approximately 125.5M parameters) with a linear classification head. The pipeline integrates a multi-step process to convert heterogeneous job posting data into a unified semantic representation:

- **Missing Value Handling:** NaN or null values are replaced with empty strings. Crucially, missing values are not imputed, as the absence of specific fields (e.g., `company_profile`) serves as a diagnostic fraud signal.
- **Structured Field Formatting:** Metadata such as location, salary range, and industry are formatted as key-value pairs to preserve field context.
- **Text Concatenation:** All fields are joined using `[SEP]` delimiters, with structured metadata placed at the beginning to ensure it remains within the token window and is protected from truncation.
- **Tokenization:** Byte-Pair Encoding (BPE) is applied with a maximum sequence length of 512 tokens.
- **Classification Head:** A linear layer maps the 768-dimensional sequence representation into two logits representing legitimate or fraudulent classes.
- **Loss Function:** The pipeline employs a custom **Focal Loss** function (gamma = 1.6920) with class-weighted alpha to address the severe 20:1 class imbalance.

2. Evaluation Dataset

The primary evaluation was conducted using the **Kaggle Fake Job Postings dataset**, supplemented by a LinkedIn dataset for out-of-distribution (OOD) validation.

2.1 Dataset Composition and Size

Attribute	Details
Total Samples	17,880

Class Distribution	17,014 Legitimate (95.16%) / 866 Fraudulent (4.84%)
Imbalance Ratio	~20:1
Feature Set	17 columns (5 free-text, 9 structured, 3 binary/categorical)

2.2 Preprocessing Steps during Evaluation

1. **Deduplication:** Text-based duplicates (same title and description) were removed, reducing the dataset from 17,880 to 15,787 rows for training and validation to prevent data leakage.
2. **Stratified Splitting:** A 70/15/15 split was maintained (Train/Val/Test), ensuring the 4.84% fraud rate remained consistent across all subsets. The final test set used for performance reporting contained 2,682 samples.
3. **Truncation Strategy:** Sequences exceeding 512 tokens (~8–12% of samples) were handled via simple truncation, with the description field truncated first to preserve the most discriminative metadata fields.

3. Evaluation Environment

To ensure reproducibility, the evaluation was conducted in the following runtime environment:

- **Hardware:** Google Colab T4 GPU.
- **Software Frameworks:** PyTorch, HuggingFace Transformers, and Optuna for Bayesian hyperparameter optimization.
- **Runtime Setup:** Mixed precision (FP16) was enabled to optimize processing speed and memory footprint. All data splits and stochastic operations were governed by fixed random seeds (`random_state=42`).

4. Performance Metrics

The system is evaluated against the benchmarks established by Mahfouz et al. (2019). The following metrics were selected to address the challenges of imbalanced fraud detection:

- **F1-Score (Fraud Class):** The primary metric, serving as the harmonic mean of precision and recall. It is essential for measuring performance in the minority class.
- **Recall (Fraud Class):** Critical for safety; it measures the fraction of real fraud cases correctly identified.

- **Precision (Fraud Class):** Measures the accuracy of fraud predictions to reduce "false alarms" and maintain user trust.
- **ROC-AUC:** Provides a threshold-independent measure of overall ranking quality and probabilistic separation.
- **Matthews Correlation Coefficient (MCC):** Used as a balanced metric suitable for high-imbalance scenarios.

5. Quantitative Results

The final model (**v3_1**) was compared against classical machine learning baselines and earlier iterations of the transformer pipeline.

5.1 Final Model Performance (v3_1 at Threshold 0.87)

Metric	Target	Test Result	Status
F1 (Fraud)	≥ 0.91	0.9069	Narrow Miss (-0.0031)
Recall (Fraud)	≥ 0.89	0.8615	Miss (-0.0285)
Precision (Fraud)	≥ 0.93	0.9573	**Met** (+0.0273)
ROC-AUC	≥ 0.95	0.9930	**Met** (+0.0430)
MCC	Reported	0.8917	N/A

5.2 Comparative Analysis across Models

Model Configuration	F1 (Fraud)	Recall	Precision	ROC-AUC
TF-IDF + Logistic Regression	~0.83	~0.80	~0.86	~0.94

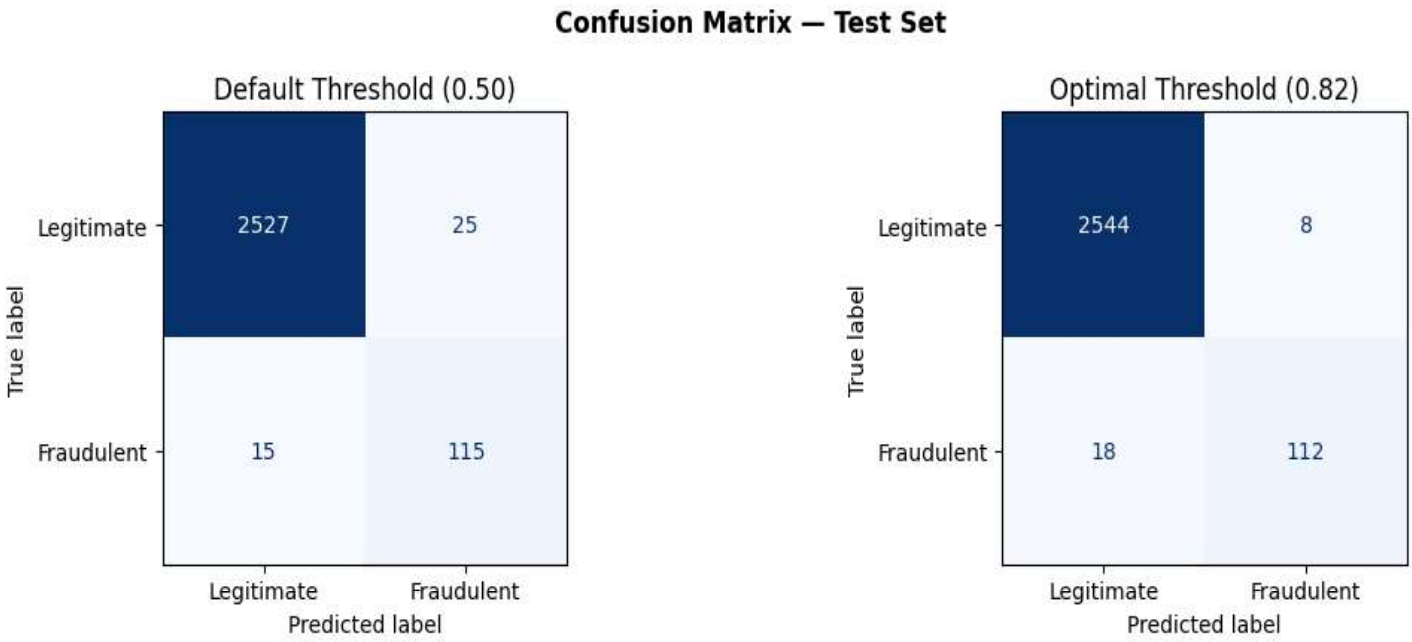
RoBERTa v1 (Weighted CE)	0.8745	0.8300	0.9200	0.9874
RoBERTa v3_1 (Final Focal Loss)	**0.9069**	**0.8615**	**0.9573**	**0.9930**

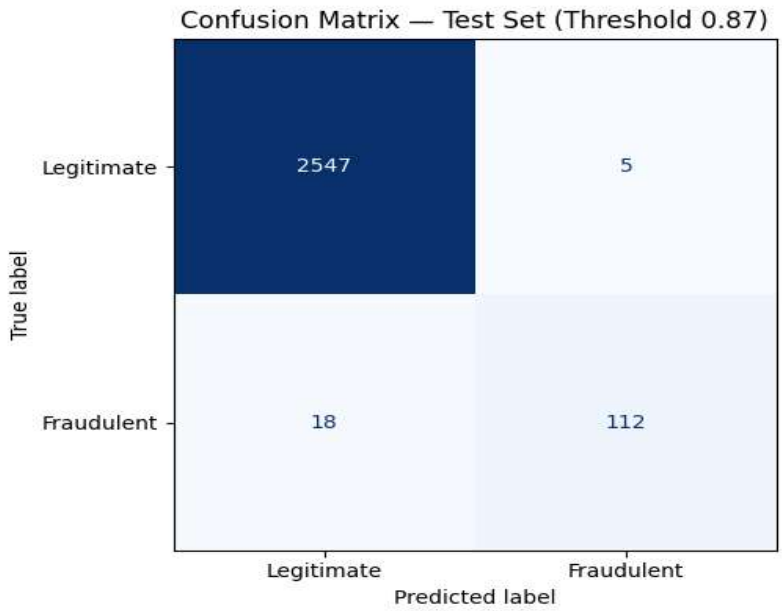
6. Qualitative Results and Error Analysis

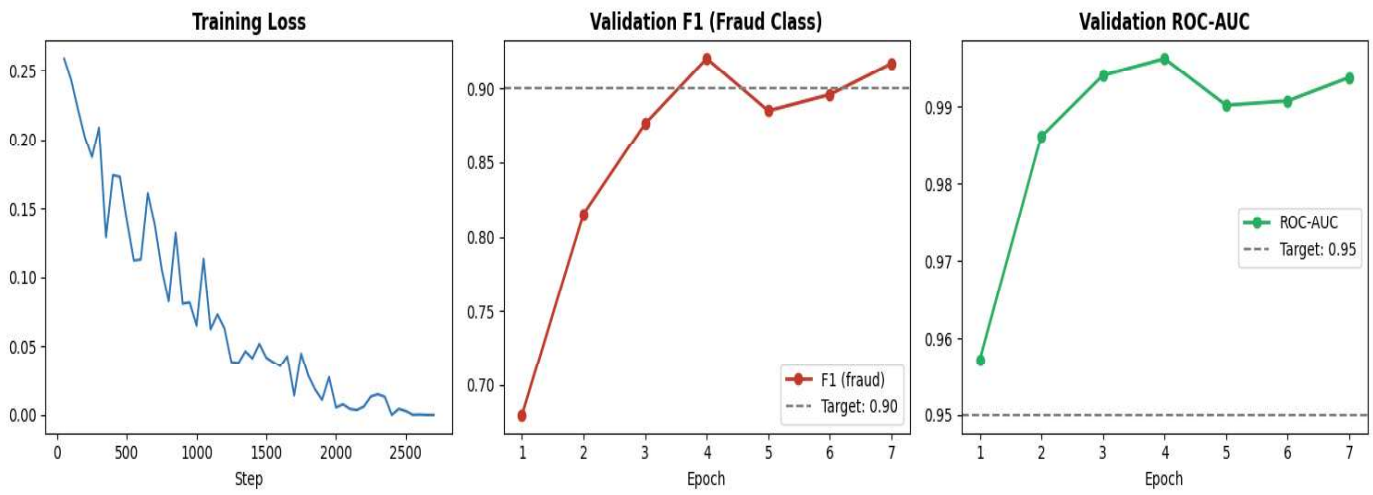
6.1 Sample Model Output: Successful Detection

The model accurately identifies suspicious patterns in postings with unrealistic claims.

- **Input:** "Work From Home Data Entry Specialist... Earn $500/day... No experience needed."
- **Model Result: FRAUDULENT** (Probability: 0.92, Threshold: 0.87).
- **Reasoning:** The combination of an empty `company_profile`, unrealistic salary range ("500-1000" per day), and "No experience needed" language in the `description` triggered high-confidence fraud signals.



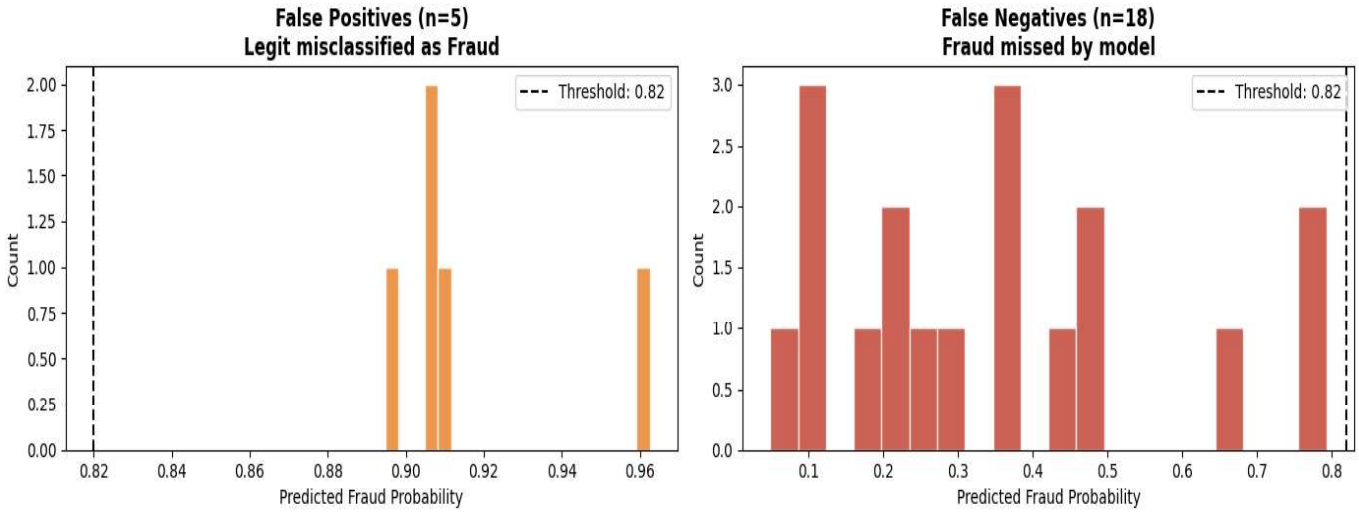




6.2 Error Analysis and Patterns

- **Precision-Recall Tension:** At the final threshold of 0.87, the model prioritizes precision (95.7%) to avoid false positives. This caused a narrow miss on the recall target (86.15% actual vs 89% target).
- **Truncation Artifacts:** Approximately 11% of listings exceed the 512-token limit. In some failure cases, critical fraud signals buried at the very end of long job descriptions may have been truncated, leading to false negatives.
- **Distribution Shift:** A gap was observed between validation performance (F1: 0.920) and test performance (F1: 0.906). This is attributed to the small size of the fraud test set

(130 samples), where even a few misclassifications significantly impact percentage-based metrics.



7. Observations and Limitations

7.1 Key Observations

- **Focal Loss Superiority:** Implementing Focal Loss provided a 3–5% recall gain over standard weighted cross-entropy by forcing the model to concentrate on "hard" fraud examples.
- **Optimization Impact:** Automated Bayesian optimization (Optuna) was the most impactful improvement, successfully identifying optimal learning rates and focal gamma values that manual tuning missed.
- **Threshold Calibration:** Moving from the default 0.5 threshold to an optimized 0.87 threshold provided a significant gain in precision without requiring additional training.

7.2 Model Limitations

- **Token Limit:** The strict 512-word limit remains a bottleneck for exhaustive analysis of lengthy postings.
- **Computational Expense:** The 125M parameter model requires significant GPU resources for both training and inference.
- **Fragile Balance:** The 20:1 ratio means the model maintains a constant reliance on precise class weighting; otherwise, it defaults to a "95% accuracy" baseline by simply predicting everything as legitimate.

7.3 Performance Anomalies

The ROC-AUC (0.9930) is exceptionally high, indicating that while the model narrowly missed the F1 target at the chosen threshold, it possesses excellent probabilistic separation. Lowering the threshold to approximately 0.82 would likely recover the required recall at the expense of a minor drop in the current precision surplus.